

RAG-Based Document Chatbot – Project Documentation

MFRP Training Plan: RAG-Based Terminal Chatbot

Project Code: MFRP-RAG-001

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Domain: AI/ML - Retrieval Augmented Generation (RAG)

Project Overview

This project implements a Retrieval-Augmented Generation (RAG) based chatbot that answers user queries using custom documents. The system retrieves relevant document content and generates responses using a Large Language Model (LLM).

Objective

- Build an intelligent document-based chatbot
- Enable semantic search over documents
- Provide accurate, context-based answers
- Implement fallback mechanism for reliability

System Architecture

User Query → Retrieval (FAISS) → Context → Prompt → LLM → Response

Key Components

1. Document Loader (PDF, TXT, DOCX)
2. Text Chunking
3. Embeddings (HuggingFace)
4. FAISS Vector Store
5. LLM Integration (Ollama + Gemini)
6. RAG Pipeline
7. Terminal Chatbot Interface

Technologies Used

Python, LangChain, FAISS, HuggingFace, Ollama, Gemini API

Execution Steps

1. Install dependencies: `pip install -r requirements.txt`
2. Install Ollama and run: `ollama pull phi3`
3. Add documents to /documents folder
4. Run: `python -m src.chatbot`

Limitations

- Scanned PDFs not supported
- Depends on document quality
- Terminal-based interface only

Future Enhancements

Implement **OCR support** to handle scanned/image-based PDFs

Develop a **web interface (Flask/Streamlit)** for better user experience

Add **conversation memory** to maintain context across queries

Integrate **hybrid search (semantic + keyword)** for improved retrieval accuracy

Use **reranking models** to refine top results before passing to LLM

Enable **API deployment** for integration with external applications

Support **multilingual queries** for wider usability

Optimize system for **large-scale document processing**

Add **user authentication and document upload system**

Fine-tune models for **domain-specific applications (e.g., healthcare, finance)**

Conclusion

This project demonstrates a complete RAG pipeline integrating document retrieval and LLM-based response generation with fallback support.